

Problem Set 3

Intermediate Macroeconomics, EC2211

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Due Wednesday, October 15 at 8 am. Worth a maximum of 1 credit for the final grade. Send one solution per group to your seminar teacher through Athena.

1 Consumption and leisure (0.5 p)

Consider a household that has N hours per month to allocate between work and leisure. Let L denote the number of hours chosen for leisure, let C denote the household's consumption and let w denote the hourly wage rate. The household cannot save or borrow. The household's utility each month is $U(C, L) = \ln C + \ln L$.

- Write down the household's budget constraint.
- Write down the household's optimization problem.
- Solve the household's optimization problem.
- How would optimal consumption and leisure choices be affected by an increase in the wage rate?

2 Consumption and savings (0.5 p)

Consider a household that lives for two periods. It has no initial wealth but receives an income Y_1 in the first period. This income can either be consumed (C_1) or saved for next period (S). The interest rate is r so saving S results in $(1+r)S$ in the second period. The household has no other income in the second period, so it then just consumes its savings. The household's life-time utility is $U(C_1, C_2) = \ln C_1 + \beta \ln C_2$ where β is the time discount factor ($0 < \beta \leq 1$).

- Write down the household's per-period budget constraints and/or the intertemporal budget constraint.
- Write down the household's optimization problem.
- Solve the household's optimization problem.
- How would optimal consumption and savings choices be affected by an increase in the interest rate in this economy?

Recommended extra exercises

Do not hand in solutions to the exercises below! Exercises marked by a * are solved in the textbook. Other exercises may be discussed in your sessions if time allows.

- Jones chapter 16: Exercises *1, 3 and 7
- Jones chapter 11: Exercise *11

Consumption and savings again

- a. Do exercise 2 again but with the following modification: Suppose that the household has income $Y_1 = 0$ in the first period and $Y_2 > 0$ in the second period. It can borrow as much as it wants (as long as it can repay!) at the interest rate r .
- b. Do exercise 2 again but with the following modification: The household's life-time utility is $U(C_1, C_2) = C_1^\theta + \beta C_2^\theta$ where $\theta = \frac{1}{2}$.